

INRC Industry Member Proposal

NeuroCUDA: Multi-Backend SNN Compiler
with Loihi 2 Hardware Validation

Submitted to: Intel Neuromorphic Research Community (INRC)

Submitted by: Quantaracore Technologies LLP

Date: June 28, 2026

Contact: Krishna Varma, Founder

Email: founder@quantaracore.in

Website: <https://quantaracore.in/neurocuda>

Member Type: Industry Member

Research Vectors: RV2 (Algorithms) + RV3 (Applications)

1. Participants

Name	Role	Affiliation	Email
Krishna Varma	Principal Investigator / Found	Quantaracore Technologies LLP	founder@quantaracore.in

2. Project Abstract

NeuroCUDA is an open-source compiler that converts trained PyTorch models to spiking neural networks and deploys them across GPU, CPU, BrainScaleS-2 analog silicon (Heidelberg), and SpiNNaker-1 digital silicon (Manchester) through one Python API call. We maintain a Loihi 2 simulator backend validated against published Loihi neuron equations (0 spike deviations per 100,000+ test vectors) but have never validated against physical Loihi 2 hardware.

This project closes that gap: deploy converted SNNs to Loihi 2 cloud hardware, validate bit-accurate spike output against our simulator, measure real energy consumption using Intel Lava energy probes, and produce honest multi-backend NeuroBench benchmarks comparing three distinct physical neuromorphic chips (Loihi 2, SpiNNaker-1, BrainScaleS-2) from one unified compiler pipeline. All Loihi 2 integration code will be released as MIT-licensed open-source.

3. Project Description

3.1 The Problem

Spiking neural network research suffers from platform fragmentation. Each neuromorphic hardware system (Loihi 2, SpiNNaker-1, BrainScaleS-2) requires its own SDK, model format, and benchmarking methodology. No open-source compiler takes one PyTorch model and deploys it to multiple physical neuromorphic chips with honest, reproducible benchmarks.

3.2 What We Already Built

Backend	Type	Status
GPU (PyTorch)	Simulator	Production. CUDA-accelerated.
CPU (PyTorch)	Simulator	Bit-exact: 0/256K spike devs vs GPU
Loihi 2 IF model	Simulator	0/100K+ spike devs vs published Loihi equations
SpiNNaker-1	Physical silicon	269K-param MLP compiles. 5000 core-hr EBRAINS quota approved
BrainScaleS-2	Physical silicon	138-neuron SNN on chip 57 (Heidelberg). Confirmed 2026-06-28
FPGA	HLS C++	NIR export pipeline working. Not yet synthesized

3.3 What is Missing - Physical Loihi 2

Our Loihi 2 simulator implements the published neuron equations correctly (0 spike deviations validated) but has never been compared to physical Loihi 2 silicon output, cannot measure real energy consumption (uses datasheet estimates: 0.08 pJ/SynOp), cannot account for fabrication variation or on-chip noise, and cannot honestly claim physical silicon per the NeuroBench hardware labeling standard.

3.4 Approach

Trained PyTorch ANN -> Stage 1: QCFS Calibration (learn per-channel thresholds, 5 epochs) -> Stage 2: IF Neuron Replacement + BPTT Fine-Tuning (binary spikes, surrogate gradients, 5 epochs) -> NIR Export (vendor-neutral graph format) -> Lava NIR Importer -> Loihi 2 Cloud Hardware (vLab) -> Validation: Spike comparison + Energy measurement + NeuroBench report.

Validation sequence (ascending complexity):

- 1. Single IF neuron: validate bit-identical spike times (simulator vs hardware)
- 2. MLP MNIST (784-256-256-10, 269K params, 97.4% SNN): full network, measure accuracy
- 3. CNN N-MNIST (event-camera, 99.88% SNN, 92% sparse): temporal input, stress-test pipeline
- 4. Multi-seed NeuroBench report: 3 seeds, full test sets, real energy from Lava probes

3.5 Comparison to Prior Work

Tool	Multi-Chip	Physical Silicon	NeuroBench	License
SNNToolBox	Partial	Partial	No	Yes
Lava-DL	Loihi only	Yes (Loihi 2)	No	Yes
sPyNNaker	SpiNNaker only	Yes	No	Yes
NIR	Format only	No	No	Yes
NeuroCUDA	3 chips	3 chips	Yes	MIT

3.6 Unique Value of Loihi 2

- Digital deterministic architecture: bit-reproducible results, unlike analog BSS-2
- Lava SDK with NIR support: clean Python API, no manual PyNN scripting needed
- Per-synapse weight programming: arbitrary matrices, unlike BSS-2 (masks only)
- Built-in energy measurement: Lava probes provide real pJ data, not datasheet estimates
- 8-bit signed weights: matches NeuroCUDA per-channel quantization pipeline exactly

3.7 Quantitative Evaluation

Metric	Target	Method
Spike deviation	0 spike time deviations	Simulator vs hardware, identical input
Accuracy gap (MLP MNIST)	$\leq 2\%$ (97.4% $\rightarrow$ $\geq 95.4\%$)	Full 10K test set, Loihi 2 vs simulator
Accuracy gap (CNN NMNIST)	$\leq 2\%$ (99.88% $\rightarrow$ $\geq 97.88\%$)	Full 10K test set, Loihi 2 vs simulator
Real energy vs estimate	Within 2x of 0.08 pJ/SynOp	Lava energy probes
NeuroBench report	Complete 3-chip table	NeuroBench standard, Nature Comms 2025

3.8 Definition of Success

A published, reproducible NeuroBench comparison table with verified numbers from three distinct physical neuromorphic chips (Loihi 2, SpiNNaker-1, BrainScaleS-2) measured by one unified open-source compiler pipeline. No other tool currently provides this.

3.9 Citations

- [1] Davies, M. et al. "Loihi: A Neuromorphic Manycore Processor with On-Chip Learning." IEEE Micro, 2018.
- [2] Orchard, G. et al. "Efficient Neuromorphic Signal Processing with Loihi 2." IEEE ISCAS, 2021.
- [3] Lava Software Framework. <https://github.com/lava-nc/lava>
- [4] NIR - Neuromorphic Intermediate Representation. <https://github.com/neuromorphs/NIR>
- [5] Yik, J. et al. "NeuroBench: Benchmarking Neuromorphic Computing." Nature Communications, 2025.
- [6] Bu, T. et al. "Optimal Quantization for SNNs via Calibrated Floor-Shift." NeurIPS, 2023.

4. Research Plan

4.1 Deliverables

#	Deliverable	Type	License
D1	Single-neuron Loihi 2 validation report	Technical report	Public
D2	Lava-based Loihi 2 backend module	Python module	MIT
D3	MLP MNIST + CNN NMNIST deployment code	Python module	MIT
D4	Multi-backend NeuroBench report (3 chips)	Public benchmark	CC-BY
D5	Tutorial: Deploy PyTorch to Loihi 2 in One Line	Blog + Seminar	Public

4.2 Personnel

Krishna Varma - Principal Investigator - Compiler pipeline, Lava integration, benchmark execution, NeuroBench reporting, documentation. Quantaracore Technologies LLP.

4.3 Milestones

Week	Milestone	Deliverable
1	INRC onboarding. Lava configured. Cloud access confirmed.	Setup verified
2	Single IF neuron on Loihi 2. Spike output validated vs simulator.	D1 complete
3-4	MLP MNIST deployed via NIR->Lava. Accuracy measured on harc	D2, D3 partial
5-6	CNN NMNIST deployed. Multi-seed benchmarks. Energy measure	D3 done, D4 partial
7-8	Lava backend complete. nc.compile(target=loihi2_lava) working.	D2 complete
9-10	NeuroBench report finalized. INRC seminar presented.	D4 complete
11-12	Tutorial published. Code merged to main. Public release.	D5 complete

4.4 Technical Tradeoffs

- Quantization: Use NeuroCUDA per-channel 8-bit quantization (matches Loihi 2 native format)
- Partitioning: MLP MNIST fits single-chip. CNN NMNIST may need 2-chip via Lava multi-chip compiler
- Encoding: Rate coding (Poisson) for MNIST. Event-driven for NMNIST. Both supported by Lava

5. Loihi Resource Needs

Project specifically targets Loihi 2 capabilities. No on-site hardware required.

Model	Params	Synapses	Chips
Single-neuron validation	<10	<10	<1
MLP MNIST (3-layer)	269,322	~269,000	1
CNN NMNIST (3-layer Conv)	147,466	~1,500,000	1-2

Cloud access pattern: Occasional interactive sessions (2-3 per week, 1-2 hours each). Batch inference runs for benchmark collection (10-20 runs per session, ~100ms each). Total estimated: approximately 100 core-hours over 12 weeks. Single-system, occasional access is sufficient.

Justification: Our simulator implements neuron equations correctly but cannot model fabrication variation, on-chip noise, or real system-level energy. Physical Loihi 2 access is required for honest physical silicon labeling per NeuroBench standard.

6. Material Deliverables to INRC

6.1 Software Contributions (MIT License)

- neurocuda/backends/loihi2_lava.py - Lava-based Loihi 2 physical silicon backend
- NIR->Lava bridge improvements (upstreamed to lava-nc if gaps found during validation)
- Energy measurement harness (standardize Lava probe output -> NeuroBench JSON format)
- Benchmark scripts and raw data (all publicly reproducible)

6.2 Documentation and Knowledge Sharing

- Deploy PyTorch to Loihi 2 in One API Call - blog post + Jupyter notebook tutorial
- Multi-chip NeuroBench report - markdown table + JSON export (3 physical chips)
- INRC Forum seminar - 30-minute presentation + Q&A
- Loihi 2 validation technical report - PDF with spike traces and energy logs

6.3 Datasets and Benchmarks

All benchmark data publicly released: Loihi 2 spike output traces (simulator vs hardware, per-neuron, per-timestep), energy measurement logs (Lava probe output per inference), NeuroBench-format JSON reports (accuracy, latency, energy, sparsity), and comparison tables across all three physical chips.

7. Intellectual Property

7.1 Background IP

NeuroCUDA is MIT-licensed open-source software. All existing code (GPU, CPU, Loihi 2 simulator, SpiNNaker-1, BrainScaleS-2 backends; QCFS converter; NIR exporter; NeuroBench reporter) is publicly available under the MIT license at <https://quantaracore.in/neurocuda>. No background IP restrictions.

7.2 IP Created Under This Project

All software, documentation, and benchmark data created under this project will be released as MIT-licensed open-source to the public domain. Quantaracore Technologies LLP does not seek to retain proprietary rights over INRC project outputs.

7.3 Agreement Type

Corporate Participation Agreement requested. Please send to founder@quantaracore.in for review and execution by Quantaracore Technologies LLP.

7.4 Funding

No Intel funding is requested. We request Loihi 2 cloud hardware access for validation only.

8. Submission Details

Field	Value
Name	Krishna Varma
Title	Founder
Company	Quantaracore Technologies LLP
Email	founder@quantaracore.in
Website	https://quantaracore.in/neurocuda
Member Type	Industry Member

Research Vectors

RV2 (Algorithms) + RV3 (Applications)

Submitted via email to inrc_interest@intel.com (online Qualtrics form reported inactive). Supporting materials: NeuroCUDA GitHub repository, multi-backend validation (5 backends, 3 simulators + 2 physical silicon confirmed), BrainScaleS-2 chip 57 confirmation (June 28, 2026), SpiNNaker-1 EBRAINS quota (5000 core-hours), NeuroBench standard compliance.

Submitted June 28, 2026 by Krishna Varma on behalf of Quantaracore Technologies LLP.